

Cloud Computing & DevOps – AWS Handbook

SESSION 11

Advanced Kubernetes & Amazon EKS

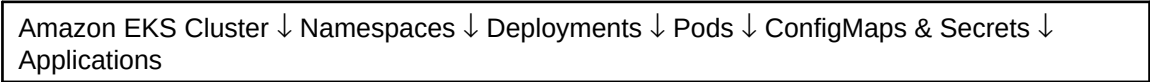
Managing Enterprise-Grade Kubernetes Workloads

Duration	2 Hours
Prerequisites	Kubernetes Fundamentals
Learning Outcome	Understand Namespaces, ConfigMaps, Secrets, Scaling, Rolling Updates, and Amazon EKS.

Why Are We Learning This?

Production Kubernetes environments require more than basic deployments. Organizations use namespaces for resource isolation, ConfigMaps for configuration management, Secrets for sensitive information, and Amazon EKS for managed Kubernetes infrastructure.

Architecture Diagram



Key Concepts

Namespace	Logical isolation of Kubernetes resources.
ConfigMap	Stores application configuration data.
Secret	Stores sensitive information securely.
Rolling Update	Updates applications with minimal downtime.
Scaling	Adjusting application capacity.
Amazon EKS	Managed Kubernetes service on AWS.

Practical Activities

- Create and manage namespaces
- Deploy applications in namespaces
- Create ConfigMaps
- Create Secrets
- Perform deployment scaling
- Perform rolling updates
- Explore Amazon EKS cluster architecture

Expected Outputs

Namespaces created successfully, ConfigMaps and Secrets configured, deployments scaled, rolling updates completed, and EKS architecture explored.

What You Should See

Paste Screenshot Here

Namespace Creation

Paste Screenshot Here

ConfigMap Configuration

Paste Screenshot Here

Secret Configuration

Paste Screenshot Here

Deployment Scaling Output

Paste Screenshot Here

Amazon EKS Dashboard

Journal Writing Template

Aim

To manage advanced Kubernetes resources and understand Amazon EKS.

Theory

Namespaces, ConfigMaps, Secrets, scaling and rolling updates are essential for managing enterprise Kubernetes workloads.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Resources were isolated using namespaces, configurations managed using ConfigMaps, and applications updated using rolling updates.

Conclusion

Advanced Kubernetes features improved application management, scalability, and operational efficiency.

Viva Preparation

1. What is a Namespace?
2. What is a ConfigMap?
3. What is a Secret?
4. What is a Rolling Update?
5. What is Scaling?
6. What is Amazon EKS?
7. Difference between self-managed Kubernetes and EKS?

Troubleshooting Box

Problem	Possible Cause
Namespace not found	Incorrect namespace reference
Configuration not applied	ConfigMap issue
Application restart failure	Deployment configuration error
Scaling not working	Resource constraints or configuration issue

Industry Relevance

Amazon EKS is widely used for running enterprise Kubernetes workloads. Namespaces, Secrets, ConfigMaps and rolling updates are standard practices in production environments.

Session Summary

Students learned advanced Kubernetes concepts, application scaling, configuration management, secure secret handling, and managed Kubernetes using Amazon EKS.